

GAURAV BOOB

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PROFESSIONAL SUMMARY

Software Engineer with 2 years of work experience building high-volume payment systems (ACH, Wire, FedWire), compliance APIs, and backend services. Skilled in Java, Spring Boot, Python, SQL, REST APIs, and AI/ML, with experience in system optimization and deep learning research. Eligible to work in Ireland without visa sponsorship.

EDUCATION

M.Sc. in Computer Science (Data Science)

Trinity College Dublin

Sept 2026 - Ongoing

B.E. in Electronics & Telecommunication

SCTR's Pune Institute of Computer Technology (PICT)

Aug 2020 – Aug 2024

GPA: 8.94 / 10.0

CORE SKILLS

Languages & Frameworks: Java, C++, Python, SQL, Spring Boot, Angular, REST APIs

Domain & Payments: FedWire, ACH, Wire Transfers, SWIFT Integration, ISO 20022 Compliance, OWASP Security

AI / ML & Computer Vision: TensorFlow, LangChain, Deep Learning, OpenCV, RAG, LangGraph, Agentic AI, LLMs

Tools & Methodologies: Git, SonarQube/SAST, Tableau, Jenkins

WORK EXPERIENCE

Software Engineer

Fidelity Information Services (FIS)

Jan 2025 – Sept 2026

Pune, India

- Built FedWire Return Search/Export APIs from scratch for US banking clients (State Street Bank) with ISO 20022 compliance layer, **reducing data truncation by 90%**.
- Led SWIFT integration across backend, database, and Angular UI; **eliminated 400+ lines of duplicate code** by consolidating shared logic into a common *AbstractPaymentCreator* base class, improving maintainability across payment modules.
- Remediated SAST/SonarQube vulnerabilities for **OWASP compliance**; built 60+ JUnit tests and Katalon automation, **cutting defect leakage 15–20%**.

Associate Software Engineer

Dragonfly Financial Technologies (Acquired by FIS)

Jul 2024 – Dec 2024

Pune, India

- Audited and remediated PII masking and sensitive data logging across **50+ backend modules**, **eliminating 40+ compliance findings** during a security-driven codebase review.
- Optimized payment platform REST APIs and database queries alongside senior DBAs, **cutting peak-load response time by 30%** across **15+ endpoints**.

Data Science Intern

Vasundhara Geo Technologies Pvt. Ltd.

Sep 2022 – Nov 2022

Pune, India

- Built computer vision pipelines using **U-Net architectures** for satellite image segmentation, achieving **87% mAP** and a **25% reduction in processing latency** across **2,000+ satellite image tiles**.

PROJECTS & RESEARCH

Lipreading for Indian Accent English

Research Project

May 2024

Demo | *Publication*

- Trained a custom CNN-BiLSTM model to interpret speech from visual cues, focusing on Indian Accent English. Achieved a **CER of 4%** and a **WER of 15%**, ranked **first position** in an international level project competition. Developed a novel Indian English accent dataset consisting of **500 videos** along with their alignment files. Research work is currently under consideration for publication at **Signal, Image and Video Processing (Springer journal)**.

Chat with Scientific Documents

Visual LLM RAG Pipeline

Mar 2024

GitHub

- Fine-tuned **Nougat Visual Transformer** for **OCR-free parsing** of mathematical equations and tables; built a **layout-aware chunking** strategy preserving section, table, and equation boundaries for accurate retrieval.
- Integrated **FAISS** for vector search and **Gemini 1.0 Pro** for document Q&A, achieving **low semantic loss** on complex academic document structures.

Classification of Diabetic Retinopathy

Deep Learning Project

Jan 2023 – May 2024

PICT Lab

- Developed hybrid CNN-LSTM models for retina image analysis, achieving **95% accuracy** in detecting retinopathy stages and macular edema - a **12-point improvement** over the baseline CNN model (83% accuracy), validated against ophthalmologist diagnoses in hospital trials.

KEY ACHIEVEMENTS & LEADERSHIP

- Winner – INC 2024 Hackathon:** Won **1st Place** in Digital Speech Processing among **200+ teams** at PICT flagship event.
- Winner – MINeD Hackathon 2024:** Secured **1st Place** among **150+ teams** (Binghamton Univ. & Nirma Univ.) for building an LLM-powered multi-format document parser.

CERTIFICATIONS

- Certifications:** Full Stack Development (Java, Spring Boot, React – Study Easy); Python for Data Science (Coursera); Applied Data Science with Python (UMich, Coursera); IBM Data Science Professional Cert.